

BlockUnitaryInterferometerSynthesisQROAM: comparing three resource-counting paths

phase\_bitsize = b = 32. ALL three paths are optimized over log\_block\_sizes  
(lambda sweep = 0..11) -- apples-to-apples comparison.

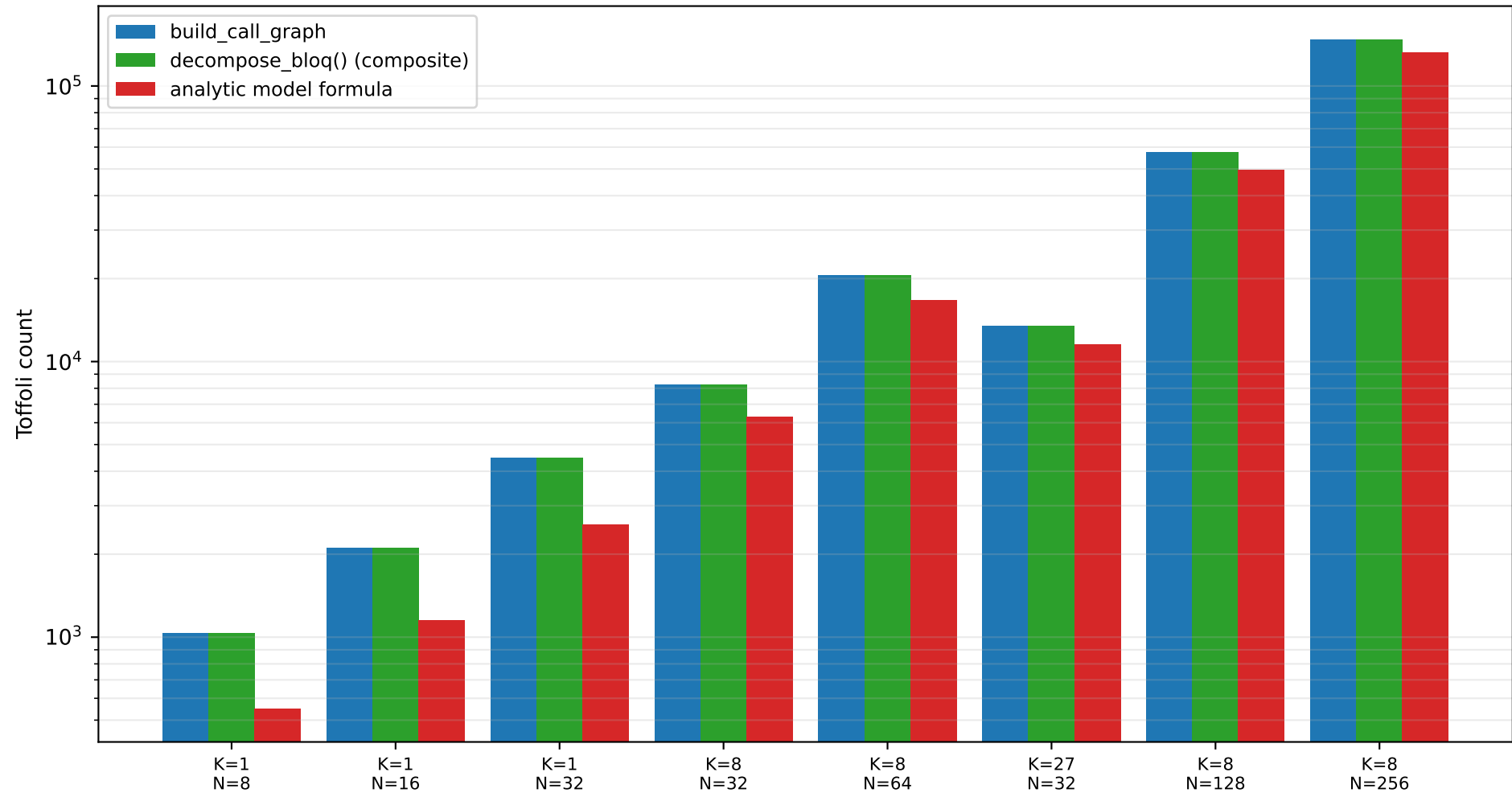
- 1) build\_call\_graph  
What get\_cost\_value(QECGatesCost()) walks by default. Returns  
Counter({phase\_layer: layer\_count, AddK +/-1: n\_odd, final\_phase: 1}).  
The walker recurses into each sub-bloq's own build\_call\_graph.
- 2) decompose\_bloq() -> composite bloq -> get\_cost\_value(QECGatesCost())  
Forces Qualtran to wire up the actual circuit (the same one defined in  
build\_composite\_bloq) and counts gates from that. This is the rigorous  
check that build\_call\_graph is a faithful summary of the wired circuit.
- 3) Analytic model formula (model\_resource\_counts.optimize\_block\_unitary\_interferometer)  
Closed-form approximation derived from the paper: a hand-rolled formula  
 $T_{\text{layer}} + (n-2)(N-1)$  shift cost +  $T_{\text{final}}$ . Drops  $O(1)$  constants and  
uses uncontrolled AddIntoPhaseGrad (the bloq uses \*controlled\*,  $\sim 2x$ ).

Headline: paths (1) and (2) agree exactly across every test case after the  
\_AbsorbedQROAMUncompute wrapper landed on the intermediate phase layer  
(per arXiv:2409.11748 the corrective sign-fix Toffolis are absorbed into  
the next layer's QROAM lookup, so 0 Toffoli is the correct local accounting).

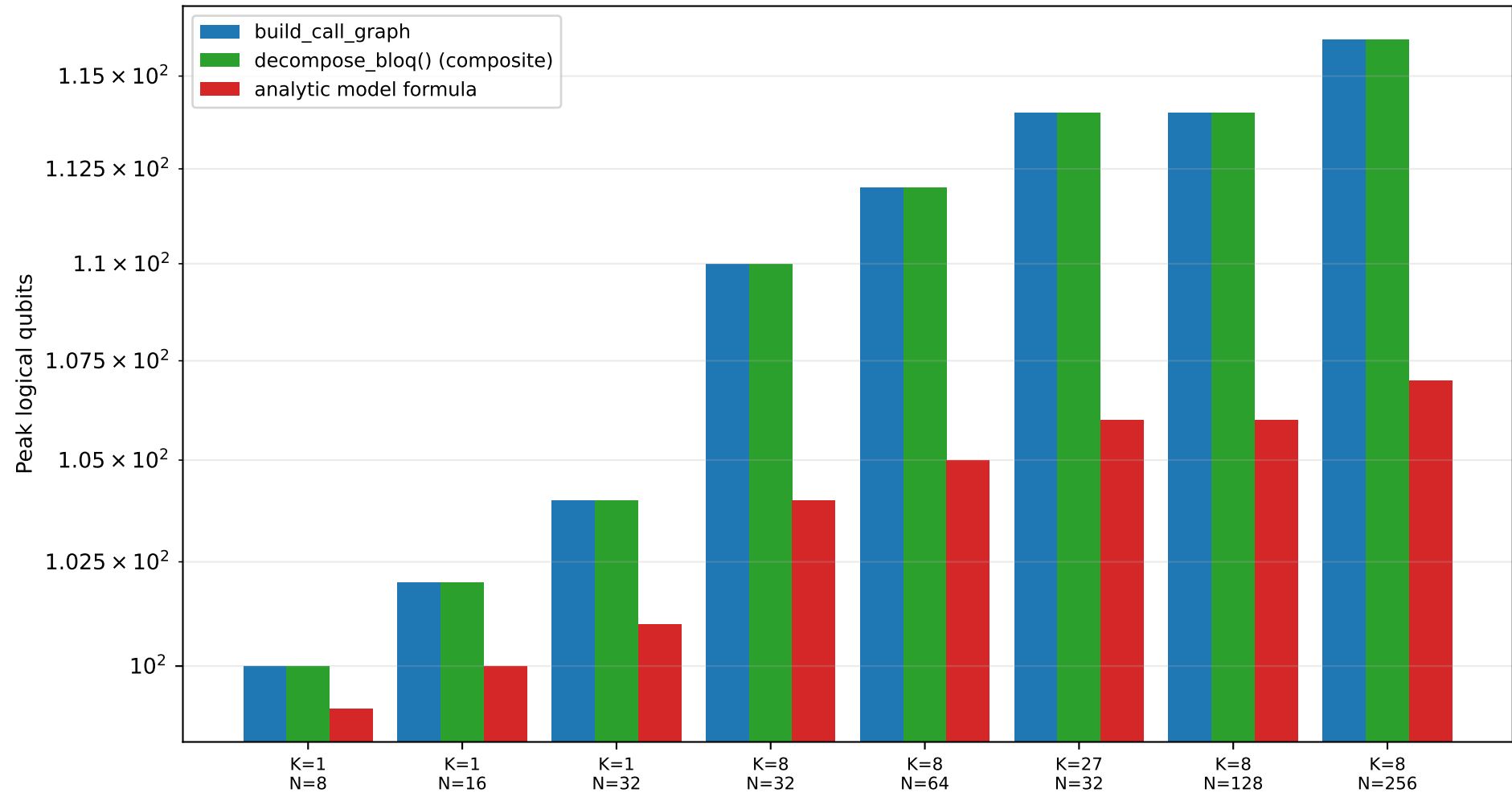
At optimal lambda the bloq is only 10-20% above the analytic model formula  
at production sizes ( $K=8$ ,  $N=128/256$ ); the residual gap is structural:  
\* controlled AddIntoPhaseGrad (2x uncontrolled) per phase layer  
\* AddK(+/-1) cyclic shifts (slightly heavier than  $(n-2)(N-1)$ )  
\* small QROAMClean constants Qualtran's internal model carries.

At small ( $K=1$ ,  $N=8$ ) the ratio stays  $\sim 1.87$  because  $M=K*N/2$  is too small to  
benefit from QROAM blocking, leaving the controlled-add overhead exposed.

Toffoli (T-opt over lambda) per path (b=32)



Peak qubits (Q-opt over lambda) per path (b=32)



# BlockUnitaryInterferometerSynthesisQROAM Toffoli counts: three paths

| K  | N   | (l1, l2) T* | call-graph T* | composite T* | model T* | T cg/model | (l1, l2) Q* | call-graph Q* | composite Q* | model Q* |
|----|-----|-------------|---------------|--------------|----------|------------|-------------|---------------|--------------|----------|
| 1  | 8   | (0, 1)      | 1,030         | 1,030        | 551      | 1.869      | (0, 1)      | 100           | 100          | 99       |
| 1  | 16  | (0, 2)      | 2,112         | 2,112        | 1,152    | 1.833      | (0, 2)      | 102           | 102          | 100      |
| 1  | 32  | (0, 2)      | 4,484         | 4,484        | 2,563    | 1.750      | (0, 2)      | 104           | 104          | 101      |
| 8  | 32  | (1, 4)      | 8,216         | 8,216        | 6,295    | 1.305      | (0, 4)      | 110           | 110          | 104      |
| 8  | 64  | (1, 4)      | 20,520        | 20,520       | 16,678   | 1.230      | (0, 4)      | 112           | 112          | 105      |
| 27 | 32  | (1, 5)      | 13,411        | 13,411       | 11,490   | 1.167      | (0, 5)      | 114           | 114          | 106      |
| 8  | 128 | (2, 5)      | 57,272        | 57,272       | 49,589   | 1.155      | (0, 5)      | 114           | 114          | 106      |
| 8  | 256 | (2, 5)      | 147,416       | 147,416      | 132,052  | 1.116      | (0, 5)      | 116           | 116          | 107      |